Faculty Initial: MDAM

Dept. of CSE

School of Engineering

Date: Jan 29, 2026 12:00 PM

Examination	: Final Examination	**Semester**	: Fall 2025
Course Code	: CSE341	**Marks**	: 30
Course Title	: Computer Networking	**Section**	: 9
Slot	: Slot - 14	**Duration**	: 2 hours

General Instructions:

- All questions must be answered.
- All the devices (mobile, smartwatch, etc) should be turned off and kept in the student's bag during the exam period.

Sl. No.	Tasks	Marks	COs	BCL*
1.	a. Prove D(E(M)) = M using RSA algorithm for public key cryptography, where E denotes encryption, D denotes decryption, and M is the message (any numerical number).	5	CO4	L-1 ...
2.	Explain how the following phenomena can happen during Three Way Handshake in TCP: a. Connection establishment b. Tearing down connection	2+2	CO2	L-4 ...
3.	Describe the three phenomena that can arise during stop-and-go in TCP.	5	CO2	L-2 ...
4.	You are tasked with allocating IP addresses for four departments within an organization. Each department requires 14 unique IP addresses for its devices. The network distributor has provided the IP address range 182.166.0.0/16. Design a subnetting scheme to meet these requirements, ensuring each department receives the necessary number of unique IP addresses. Provide the subnet details, including the network addresses, usable host ranges, and broadcast addresses for each department.	5	CO3	L-3 ...
5.	Describe how receivers interact in the sliding window protocol.	4	CO4	L-2 ...
6.	Consider the following condition for GB-4: The total number of packets is 10 that are to be sent. What will	3+4	CO4	L-2 ...

*Bloom's Cognitive Level

	happen for a given window on the following scenario- a. If acknowledgement of the 2nd packet is lost. b. If acknowledgements of all the packets are lost.			

*Bloom's Cognitive Level